



Binary numbers

NUMBERS ARE COMMONLY WRITTEN USING THE DECIMAL SYSTEM, BUT NUMBERS CAN BE WRITTEN IN ANY NUMBER BASE.

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What is a binary number?

The decimal system uses the digits 0 through 9, while the binary system, also known as base 2, uses only two digits—0 and 1. Binary numbers should not be thought of in the same way as decimal numbers. For example, 10 is said as “ten” in the decimal system but must be said as “one zero” in the binary system. This is because the value of each “place” is different in decimal and binary.

Decimal numbers



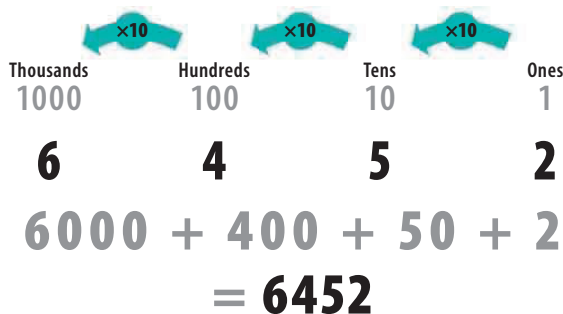
Binary numbers



a single binary digit is called a “bit,” which is short for binary digit

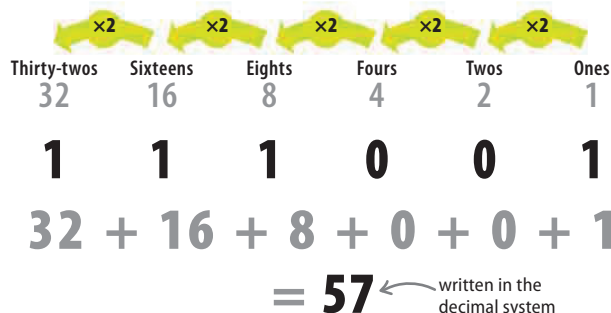
Counting in the decimal system

When using the decimal system for column sums, numbers are written from right to left (from lowest to highest). Each column is worth ten times more than the column to the right of it. The decimal number system is also known as base 10.



Counting in binary

Each column in the binary system is worth two times more than the column to the right of it and, as in the decimal system, 0 represents zero value. A similar system of headings may be used with binary numbers but only two symbols are used (0 and 1).



Decimal	Binary
0	0 0
1	1 1 one
2	1 0 1 two
3	1 1 1 two + 1 one
4	1 0 0 1 four
5	1 0 1 1 four + 1 one
6	1 1 0 1 four + 1 two
7	1 1 1 1 four + 1 two + 1 one
8	1 0 0 0 1 eight
9	1 0 0 1 1 eight + 1 one
10	1 0 1 0 1 eight + 1 two
11	1 0 1 1 1 eight + 1 two + 1 one
12	1 1 0 0 1 eight + 1 four
13	1 1 0 1 1 eight + 1 four + 1 one
14	1 1 1 0 1 eight + 1 four + 1 two
15	1 1 1 1 1 eight + 1 four + 1 two + 1 one
16	1 0 0 0 0 1 sixteen
17	1 0 0 0 1 1 sixteen + 1 one
18	1 0 0 1 0 1 sixteen + 1 two
19	1 0 0 1 1 1 sixteen + 1 two + 1 one
20	1 0 1 0 0 1 sixteen + 1 four
50	1 1 0 0 1 0 1 thirty-two + 1 sixteen + 1 two
100	1 1 0 0 1 0 0 1 sixty-four + 1 thirty-two + 1 four